

AI for Financial Practitioners

Lecture 1 · Part I

From Models to Agentic Organization

A pedagogical monograph

How models become working systems through tools, skills, memory, agents and the deliberate organization of intelligent work

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ABSTRACT

This monograph develops an organizational view of agentic AI. It distinguishes a model from a working system and then separates tools, reusable skills and memory as different extensions of capability. An agent is defined not by conversational style but by a bounded responsibility, an observable state, permitted actions and an escalation path. The discussion then moves from individual agents to coordinated specialist arrangements, showing that multi-agent design is a problem of organizational architecture: decomposition, communication, verification, conflict resolution and accountability must be designed together.

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LECTURE 2 · PART I

From Models to Agentic Organization

A visual guide to the key ideas of the monograph

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1 FROM A MODEL TO A WORKING SYSTEM
A model provides a capability; an institution requires an architecture that turns capability into sustained work.

Model	System	Institutional Work
Intelligence and reasoning capability	Connected capabilities	Sustained, governed and valuable outcomes
Generates responses based on data and patterns	Models, tools, data, memory, procedures and people	Reliable, repeatable and integrated into decision-making

2 TOOLS, SKILLS AND MEMORY ARE DIFFERENT
Each component plays a distinct role in turning model capability into useful and reliable work.

Tools	Skills	Memory
Operations in the external world	Reusable procedures	Persistent, qualified information
Data access and retrieval	Prompting patterns	Conversation history
Web and APIs	Analytical methods	Institutional knowledge
Document processing	Domain procedures	Domain reference data
Calculations and simulations	Decision frameworks	Experience and feedback
Transactions and workflow systems	Structured outputs	User preferences and working context
Allow the system to act on or interrogate the external world.	Encode institutional knowledge into repeatable procedures.	Preserve relevant information and experience across interactions.

3 THE AGENT AS A UNIT OF RESPONSIBILITY
An agent combines objectives, context, capabilities, tools, memory and constraints to pursue a bounded responsibility over time.

AGENT
A unit of responsibility

An agent is more than a language model.
It is a configured system with a role, resources and operating boundaries.

4 FROM INDIVIDUAL AGENTS TO MULTI-AGENT ORGANIZATION
As responsibilities become more complex, work can be distributed among specialist agents.

5 AI ARCHITECTURE RESEMBLES ORGANIZATIONAL DESIGN
The key question is no longer which model to use, but how to divide responsibilities among humans, models, tools and agents.

Design Dimensions

- What responsibilities should be assigned to humans, models and agents?
- How should the work be divided?
- How do they interact, coordinate and escalate?
- What authority, permissions and constraints apply to each role?
- How is performance monitored and improved over time?

From Technical to Organizational Thinking

From Which model?	→	To Which organization?
From Higher capability?	→	To Clearer responsibility?
From Isolated performance?	→	To End-to-end outcomes?
From Model configuration?	→	To Governance and oversight?
From Technical deployment?	→	To Institutional design?

"From model capability to organized institutional work."

Lecture 1 gave us an operating cycle: perceive the world, form a judgment, plan an action, execute within authority and learn from what follows. Lecture 2 asks who, or what, performs those responsibilities. The answer begins with a model, but it does not end there. The central lesson of this first half is that institutional intelligence is organized intelligence.

Introduction: when intelligence acquires an organization

MAIN IDEA

Once models are connected to tools, memory and recurring responsibilities, AI design becomes a problem of organizing intelligent work.

A useful way to enter Lecture 2 is to return to the research service that has accompanied us since the beginning of the course. At first the service looked almost like a conversation. An analyst supplied a filing, a capable language model read it, and a memorandum appeared. Lecture 1 taught us to look behind that apparently simple exchange. The memorandum depended on evidence, context, calculations, interpretations, predictions and decisions about what could safely be said. We then imagined the same capability operating continuously across hundreds of companies and discovered that intelligence had to be organized as a governed process.

Now a different question arises. Who performs the work inside that process? If a filing arrives, something must obtain it. Something must identify the company and reporting period. Something must calculate the ratios. Something must interpret changes. Something must compare the new evidence with what the institution already believes. Something must decide whether an inconsistency deserves escalation. A person could perform every one of these tasks, and historically people often did. A single model could also be asked to imitate the entire chain. But neither arrangement is necessarily the best way to design the service.

Organizations exist because complex work benefits from a division of responsibility. A bank does not normally ask the same person to originate a loan, approve it, book it, settle it and audit it. An asset manager distinguishes research, portfolio construction, trading, risk and operations even though all of them contribute to one investment process. The separation is not merely bureaucratic. Different responsibilities require different information, different expertise, different permissions and different forms of evidence.

Agentic AI brings this organizational question into software. A model can be surrounded by tools, reusable procedures, memory and operating instructions. It can be given a bounded objective and allowed to choose among permitted steps. Several such units can be arranged so that one specializes in evidence, another in calculation, another in interpretation and another in review. At that point we are no longer discussing a model in isolation. We are designing an artificial organization.

The phrase can sound futuristic, but the underlying idea is practical. We are deciding how work should be decomposed, where discretion is useful, what information should persist, how responsibilities should meet, and where human judgment should remain decisive. The purpose of Part I is to build a vocabulary for making those choices without mystifying them.

1 From a model to a working system

MAIN IDEA

A model becomes a working system only when surrounding components provide evidence, operations, state, controls and a route to completion.

A model is a capability. It maps an input into an output. That description is deliberately broad enough to include a regression, a classifier, a forecasting model and a large language model. The important point is that the model does not, by itself, define the institutional process in which its output will be used.

Consider a credit committee. A model may estimate the probability that a borrower defaults. The committee still needs verified financial statements, an exposure calculation, collateral information, covenant analysis, a record of prior decisions and a clear statement of the authority under which the loan may be approved. The model contributes something important, but the credit process cannot be reduced to the model.

Generative AI sometimes obscures this distinction because the interface is so encompassing. A language model can read a document, discuss it, write code, summarize a policy and propose an action within one conversation. The experience creates an impression of a single intelligent object doing everything. Yet behind a dependable implementation we usually want sharper boundaries. The document should come from an authorized source. Numerical calculations should use controlled definitions. Persistent facts should enter memory only under specified conditions. Actions should

be performed through tools whose permissions can be limited and logged.

This is why the phrase *AI system* matters. A system is an arrangement of components whose interactions produce the behavior we care about. The model may be the most cognitively flexible component, but the surrounding architecture determines what information reaches it, what operations it can invoke, what state survives, what happens after it produces an answer and how failure is detected.

The distinction becomes vivid when we imagine replacing one model with another. If the architecture is well designed, a stronger model may improve interpretation without changing the rules governing access to the portfolio database. The calculation engine can remain the same. The approval workflow can remain the same. The audit record can remain the same. We have improved a capability without rebuilding the institution around it.

This modularity is not merely an engineering preference. It is a governance advantage. Responsibilities that are separated can be evaluated separately. If the final memorandum is wrong, we can ask whether the source was stale, the entity was misidentified, the calculation was incorrect, the interpretation was weak or the review failed. When every responsibility is hidden inside one prompt, failure becomes harder to diagnose.

There is a broader managerial lesson here. The question “Which model should we buy?” is usually too early. We first need to understand the work. What is the objective? What evidence enters? Which transformations must be exact? Where is judgment useful? Which actions have consequences? Once the task is decomposed, model choice becomes one architectural decision among several rather than the entire strategy.

A financial institution therefore moves from model selection to system design. The model remains important, but it becomes one participant in a larger arrangement. That is the first conceptual step toward agentic organization.

2 Tools, skills and memory: three different extensions of intelligence

MAIN IDEA

Tools perform operations, skills encode repeatable procedures and memory preserves qualified state; each extension requires distinct controls.

Once we stop treating the model as the whole system, three surrounding capabilities

become especially important: tools, skills and memory. They are sometimes grouped together in casual conversation, but they solve different problems.

A tool extends what the system can do. Imagine an analyst who can reason very well but has no calculator, no terminal and no access to the filing archive. The analyst may understand a question but be unable to perform the operation required to answer it. A tool changes that. It may be a database query, a pricing library, a spreadsheet function, a search service, a code interpreter or an interface to an internal workflow. The tool connects reasoning to an operation.

This distinction is valuable because a model need not imitate a capability that already exists in a more reliable form. If we need to calculate the present value of a known sequence of cash flows, we can use deterministic code. If we need the current approved position of a portfolio, we can query the system of record. If we need to retrieve the executed version of a contract, we can use a controlled document repository. The language model can decide that the operation is relevant, interpret the result and explain it. It does not need to hallucinate the operation from memory.

A skill is different. In the vocabulary of this course, a skill is a reusable procedure for accomplishing a type of work. It is closer to a professional method than to an instrument. “Review this issuer for refinancing risk” can be represented as a skill that specifies the evidence to collect, the ratios to calculate, the maturities to inspect, the scenarios to consider and the conditions that trigger escalation. The skill may use several tools and perhaps several models. Its purpose is to give recurring work procedural structure.

This is familiar in finance. Institutions are full of skills even when they do not use that word. A due-diligence checklist, a valuation protocol, a trade-surveillance procedure and a credit-review methodology are reusable ways of performing work. Encoding such procedures for AI does not mean freezing professional judgment. It means making the stable parts of the process explicit enough that judgment can be exercised where it is actually needed.

Memory solves a third problem: continuity. Without memory, every interaction begins as if the institution had never seen the case before. A research service needs to know which filing was last reviewed, what assumptions were approved, which inconsistency remains unresolved and whether a warning has already been escalated. Memory allows relevant state to persist.

But memory is not automatically knowledge. A persistent statement can be wrong. A once-valid assumption can become stale. A preliminary interpretation can accidentally

acquire the status of an established fact. This is why institutional memory needs qualification. A useful memory record should tell us what the information is, where it came from, when it was observed, whether it was verified, who or what approved it and when it should be reconsidered.

Think of a simple example. An agent reads that a company intends to refinance a bond before maturity. Should that statement become permanent memory? Perhaps it should be stored as a management intention with a source and date, not as the fact that refinancing has occurred. Months later the system should be able to distinguish the old intention from a completed transaction. The difference is economically material.

Tools, skills and memory therefore enlarge intelligence in three distinct directions. Tools enlarge the action space. Skills enlarge procedural competence. Memory enlarges continuity through time. Combining them with a model creates something much closer to a working professional role. It also creates new governance questions, because the system can now reach external resources, follow procedures and carry information from one episode into the next.

3 The agent as a unit of responsibility

MAIN IDEA

An institutional agent is best defined by a bounded responsibility, observable state, permitted actions, completion criteria and escalation path.

The word *agent* is used so broadly that it can become almost meaningless. For this course, the most useful definition is organizational rather than anthropomorphic. An agent is a bounded unit of intelligent work that receives an objective, has access to specified information and capabilities, can choose among permitted steps, and produces an output under explicit constraints.

The key word is *bounded*. We do not need to imagine a digital employee with a personality. We need to identify a responsibility for which some discretion is useful. A filing agent might determine which sections of a newly published report are relevant to a credit-monitoring assignment. A calculation agent might decide which approved ratios are applicable to the issuer and call deterministic tools to compute them. A review agent might compare the draft conclusion with the underlying evidence and identify unsupported claims.

Why introduce an agent at all? Because some tasks cannot be specified efficiently

as a fixed sequence of rules. The path depends on what is discovered. A filing may contain a new debt instrument that requires a different analysis. A data source may be unavailable, forcing the system to seek an approved alternative. Two documents may disagree, requiring further investigation. An agent can exercise bounded discretion over the route while remaining constrained by the objective and permitted actions.

This is analogous to delegation in a human organization. A junior analyst is not given a script for every possible sentence in a filing. The analyst receives an assignment, access to certain systems, professional procedures, a deadline and a supervisor. Within those boundaries, the analyst decides where to look and what deserves attention. The quality of the organization depends partly on whether the assignment and boundaries are well designed.

An artificial agent should be described with the same seriousness. What is its objective? Which tools may it use? Which memory may it read or write? What evidence must accompany its conclusion? How much time or computational budget may it consume? When must it stop? What uncertainty requires escalation? Who receives its output? These questions transform the vague instruction “use an agent” into a design that can be examined.

The agent also provides a useful unit for accountability. Suppose a monitoring system misses a deterioration in liquidity. If the evidence-acquisition responsibility was explicit, we can ask whether the relevant filing was obtained. If the interpretation responsibility was explicit, we can ask whether the evidence was understood. If the review responsibility was explicit, we can ask why the omission was not detected. Accountability does not require pretending that software is morally responsible. It requires making the institutional allocation of responsibility visible enough that humans can govern it.

There is a temptation to give every activity its own agent because the resulting diagram looks sophisticated. That is not the objective. An agent introduces discretion, coordination cost and another place where behavior can diverge from expectation. If deterministic code can perform a responsibility transparently, an agent may add little. If two responsibilities use the same evidence, the same rules and the same authority, splitting them may create ceremony rather than control.

The right question is therefore not “How many agents can we deploy?” It is “Where does bounded discretion create value?” This keeps the architecture connected to the economics of the work.

4 From individual agents to multi-agent organization

MAIN IDEA

Multiple agents create value through meaningful specialization and independent information, not through the mere multiplication of artificial voices.

Once we have one agent, it is natural to imagine several. The attraction is obvious. Finance already depends on specialization. A transaction may require legal, accounting, tax, market, risk and operational perspectives. An investment committee benefits from analysts who examine a company through different lenses. A regulator separates supervision, enforcement, policy and data responsibilities. Why should an artificial system not also divide work among specialists?

Sometimes it should. A multi-agent arrangement can make heterogeneous responsibilities explicit. One agent may focus on source evidence, another on financial calculations, another on qualitative interpretation and another on review. They can operate sequentially, in parallel or through a supervisor that allocates work. The structure can make a complex process easier to inspect because each contribution has a defined purpose.

But multiple agents do not create reliability merely by voting. If five agents read the same erroneous summary, five agreeing answers may simply reproduce one upstream failure. Diversity of labels is not the same as diversity of evidence. A reviewer is useful when it has a concrete checking method: access to original documents, an independent calculation, a different source or a defined set of failure tests.

This point is particularly important because conversational agents can appear to disagree and debate in ways that resemble human committees. The performance may be intellectually engaging, but institutional design requires more than theatrical pluralism. We need to know what information each agent possesses, what responsibility it carries and why its contribution adds information or control.

Coordination then becomes a first-class problem. If one agent extracts a debt maturity and another calculates liquidity, they must refer to the same issuer, date, currency and definition. A handoff that loses context can destroy the benefit of specialization. The receiving agent should know not only the value but its provenance and status. In this sense, the interfaces between agents are as important as the agents themselves.

Disagreement is also information. Suppose an interpretation agent concludes that leverage is deteriorating while a reviewer argues that the apparent increase reflects a

temporary acquisition bridge. The system should not necessarily force an artificial consensus. A well-designed architecture can preserve the disagreement, identify the assumptions that produce it and escalate the issue to a human decision-maker. Institutions often need disciplined dissent more than premature unanimity.

There is also a question of hierarchy. Some arrangements use a supervisor agent that decomposes the assignment, delegates tasks and integrates results. Others use a more decentralized protocol. Neither is inherently superior. The choice depends on the task. A supervisor can simplify coordination but can also become a single point of failure. Decentralized interaction can increase robustness in some settings while making accountability and stopping conditions harder to manage.

The central pedagogical point is modest: multi-agent systems are organizational designs. Their value comes from a useful division of labor, informative independence and controlled coordination, not from the number of artificial voices in the room.

5 AI architecture as organizational design

MAIN IDEA

Agentic architecture programs a division of labor, making responsibility, handoffs, concentration risk and human participation design choices.

We can now make the larger move. Once models have tools, procedures, memory and bounded responsibilities, AI architecture begins to resemble organizational design. The resemblance is not metaphorical decoration. The same questions appear in both domains: who does what, with which information, under what authority, through which handoffs, subject to which review, and with what mechanism for resolving exceptions?

Imagine a bank building an AI-supported credit process. One design might place a powerful model at the center and allow it to perform nearly every task. Another might use deterministic services for financial spreading, a retrieval system for policy and legal documents, an agent for qualitative analysis, a separate agent for challenge, and a human credit officer for approval. The second architecture may look more complicated, but its responsibilities are easier to inspect. Whether the additional complexity is justified depends on the economics and risks of the process.

Organizational thinking also clarifies the role of humans. The arrival of agents does not create a simple contest between human work and machine work. It creates a new

allocation problem. Some responsibilities are attractive candidates for automation because they are repetitive, observable and reversible. Some are attractive candidates for AI assistance because they require synthesis but benefit from human judgment. Some should remain human decisions because authority, accountability, fiduciary responsibility or contextual understanding makes human participation essential.

The design can therefore be expressed as a responsibility map. For every significant step we ask what capability performs the work, what evidence it uses, what it may change and who owns the consequence. This is more informative than a technology diagram filled with product names. It tells the institution how intelligence is actually organized.

The same map reveals concentration risk. If every specialist depends on the same model, the same retrieval index and the same mistaken memory, apparent organizational diversity may hide a common failure mode. If every agent can write to the same persistent record, one erroneous conclusion may contaminate the institution's future reasoning. If the supervisor can override every control, the hierarchy may be efficient but fragile. Organizational design forces us to look at these dependencies.

It also helps us understand scale. A single analyst using an assistant can compensate informally for ambiguity. A thousand automated cases cannot depend on the analyst noticing every anomaly. As scale increases, responsibilities and interfaces must become more explicit. The organization must know what normal completion looks like, what evidence accompanies it and what exceptions stop the process.

This is the deeper reason that agentic AI matters for executives. The technology is not only another analytical instrument. It makes portions of organizational structure programmable. Procedures, responsibilities, memory, coordination and escalation can increasingly be represented in software. That creates enormous productive potential, but it also means that poor organizational choices can be automated with unprecedented speed.

6 The agent contract: responsibility before personality

MAIN IDEA

A clear agent contract specifies purpose, inputs, tools, outputs, constraints, refusal conditions and accountable ownership before conversational style.

The word "agent" often evokes a conversational persona. For institutional design,

personality is secondary. The useful starting point is an agent contract: a precise description of the responsibility assigned to the component and the conditions under which that responsibility is complete. The contract identifies purpose, inputs, permitted operations, outputs, state, limits, escalation and the owner of the downstream consequence.

Consider an evidence agent in an investment-research workflow. Its purpose is to assemble authorized evidence relevant to a defined question. Its inputs include an issuer identity, date, question and access context. It may search approved repositories and rank passages. It may not infer missing financial values, use public web sources without authorization or write to the approved investment record. Its output contains passages, document identifiers, effective dates and retrieval confidence. Completion means that the evidence package satisfies coverage rules or that the agent explicitly reports why it cannot.

This contract distinguishes the agent from a general assistant. The agent possesses bounded discretion about how to search, query expansion and ranking, but not about the institutional purpose of the search. It can choose a path without choosing its mandate. Russell and Norvig's classical definition of an agent emphasizes perception and action in an environment [1]; the institutional contract adds ownership, evidence and authority because a financially useful environment is also an organizational and legal environment.

Contracts improve replaceability. If the output schema and acceptance tests are stable, an institution can compare alternative models or implementations without redesigning the whole workflow. They also improve incident analysis. When an output fails, the team can ask whether the contract was wrong, the inputs were invalid, the agent violated a constraint or the receiving component misunderstood the handoff.

The contract should include negative capability: what the agent must refuse to do. It may abstain when the entity is ambiguous, the source is stale or the task exceeds its authorization. Refusal is not a failure when it preserves the boundary. The architecture should distinguish a successful abstention from an operational error and route each appropriately.

Finally, the contract names a human or organizational owner. Assigning work to software does not create accountability inside the software. Someone owns the definition, approves material changes, reviews performance and decides how exceptions affect the business process. The clearer the contract, the less likely the institution is to confuse a model's fluent initiative with organizational authority.

7 Tools, skills and memory as controlled assets

MAIN IDEA

Institutional capability depends on governing the lifecycle, permissions, provenance, versioning and correction of tools, procedures and stored knowledge.

Tools, skills and memory extend an agent in different directions. A tool changes the world or obtains a result from an external capability. A skill packages a repeatable procedure. Memory preserves qualified state across steps or episodes. Treating the three as controlled assets helps the institution manage their distinct risks.

A tool should have a narrow interface and an explicit permission model. A market-data tool might accept an instrument identifier, fields and an observation time. A calculation tool might accept controlled inputs and return both a value and a trace of the formula. A communication tool should distinguish preparing a message from sending it. Each invocation needs an identity, purpose, parameter validation, timeout behavior and record of the result.

A skill converts tacit craft into an executable procedure. A covenant-review skill may direct the agent to identify the governing agreement, resolve amendments, locate defined terms, extract numerator and denominator, reconcile the calculation and flag interpretive uncertainty. The skill does not guarantee correctness. It creates a stable object that can be reviewed, versioned and tested. When an experienced analyst improves the procedure, the institution can evaluate the change against prior cases rather than depend on informal prompt editing.

Memory requires the strongest semantic discipline. Working memory contains temporary context for the current assignment. Episodic memory records prior cases or interactions. Semantic memory preserves approved facts, definitions or organizational knowledge. Procedural memory may preserve how a task is performed. Mixing these categories can create subtle failures. A tentative analyst comment should not silently become an approved fact; a procedure that worked for one jurisdiction should not become a universal rule.

Memory entries therefore need provenance, scope, confidence, owner, creation time and expiry or review conditions. They should be retrieved according to the current user and purpose, not merely because they are semantically similar. Sensitive content may be permissible in a case workspace and impermissible in a broadly shared long-term memory. Deletion and correction must propagate to indexes and derived state

where policy requires it.

The asset perspective changes procurement as well. The institution should not ask only whether a platform “supports tools and memory.” It should ask whether tools can be restricted independently, whether skill versions can be pinned and tested, whether memory can be inspected and corrected, and whether a complete action can be reconstructed. Capabilities become institutional when their lifecycle can be governed.

8 Decomposition and the economics of coordination

MAIN IDEA

Work should be divided only where specialization or control exceeds the cost of handoffs, shared state, latency and organizational complexity.

Dividing work among agents can improve specialization, parallelism and inspectability, but division creates coordination cost. Malone and Crowston define coordination as managing dependencies among activities [3]. That formulation is directly applicable to agentic systems. The architecture should be chosen by studying dependencies, not by assuming that more specialists imply more intelligence.

Three dependency types recur. A flow dependency exists when one agent’s output becomes another’s input. A shared-state dependency exists when several agents read or update the same case record. A resource dependency exists when agents compete for a limited tool, data quota, approval queue or time window. Each type requires a different control: schemas and validation for flows, concurrency and ownership rules for state, and scheduling or budgets for resources.

Decomposition is useful when responsibilities require different evidence, tools or evaluation. Numerical spreading and qualitative strategy analysis are natural candidates because one benefits from deterministic reconciliation and the other from interpretive breadth. Decomposition is less useful when a task is small and tightly coupled. Splitting every paragraph among agents can produce costly handoffs, inconsistent context and little independent information.

Organizational theory offers another caution. Herbert Simon described complex systems as nearly decomposable when interactions within components are stronger than interactions between them [2]. Agent boundaries work best when they follow

such structure. If every decision requires constant negotiation across every specialist, the chosen modules may not reflect the task's real organization.

The economics should include model calls, tool calls, latency, failure recovery and human review. A five-agent committee that improves a benchmark by a small amount may reduce business value if it triples completion time and creates opaque disagreement. Conversely, a specialist verifier may be worth substantial cost when it independently checks a consequential calculation. The relevant measure is marginal information or control per unit of coordination cost.

A practical design begins with the minimum organization. Use one bounded agent and controlled tools where that is sufficient. Add a specialist when it contributes a distinct capability or independent check. Add a supervisor when allocation and integration are themselves material responsibilities. Preserve the reason for each role so that the organization can later be simplified if its contribution is not observed.

9 Verification, independence and disciplined disagreement

MAIN IDEA

Review adds information when it uses independent evidence or methods and preserves material disagreement instead of manufacturing consensus.

Multi-agent review is valuable only when the reviewer has a basis for detecting error. Asking a second instance of the same model to “check” an answer can catch some inconsistencies, but it can also reproduce the same misconception with greater confidence. Informative independence comes from different evidence, methods or incentives.

A calculation verifier should recompute from source data rather than read the first agent's conclusion. A citation verifier should open the cited passage and test whether it entails the claim. A policy verifier should query the approved rule set and version. A red-team agent may search specifically for disconfirming evidence or violated assumptions. These roles possess concrete checking procedures.

Wooldridge's treatment of multi-agent systems emphasizes interaction, coordination and strategic behavior among autonomous components [4]. Inside a firm, the strategic problem is designed rather than naturally given. The institution chooses what each

agent sees, what it is rewarded for and how disagreement is resolved. If every agent is optimized to complete the case quickly, they may converge prematurely. Giving one role an explicit mandate to identify missing evidence changes the information produced by the organization.

Disagreement should be represented at the level of claims and assumptions. Suppose a fundamental agent expects margins to recover while a sector agent expects persistent pressure. The integrator should not average the prose. It should identify whether they disagree about volume, price, costs, timing or management credibility. Scenario analysis can then preserve both views and reveal the variables that would discriminate between them.

Escalation rules should reflect materiality. Minor formatting differences can be resolved automatically. A discrepancy in a covenant ratio, policy interpretation or issuer identity should stop dependent action. The system should record which agent raised the issue, the evidence involved and the final resolution. This creates data for evaluating whether the review organization catches important failures or merely generates noise.

Human committees face similar challenges, but artificial arrangements can amplify them because agents operate quickly and produce large volumes of plausible material. A disciplined protocol is therefore more important, not less. Turn-taking, schemas, stopping conditions and decision rights prevent an open-ended simulation of debate. The goal is not to imitate a meeting. It is to produce the evidence and challenge required by the responsibility.

10 Worked design: an agentic investment memorandum

MAIN IDEA

A robust memorandum workflow separates evidence, financial analysis, market context, challenge, integration and human approval through explicit interfaces.

An asset manager wants a weekly memorandum for each issuer on a credit watch list. The current process combines document collection, financial updates, market observations, qualitative assessment and review. The agentic design begins with a case manager that creates a case identifier, issuer graph, observation date and required output. It does not perform every analysis; it maintains the state of the assignment.

An evidence agent collects authorized filings, transcripts, rating reports and internal

notes. A financial agent calls controlled tools to update debt, liquidity, coverage and maturity measures. A market agent obtains prices, spreads and comparable observations with timestamped sources. A qualitative agent examines business developments, management statements and sector conditions. Each writes to a separate part of the case record using a defined schema.

A challenge agent receives the source packages and claims but not merely the polished draft. It checks citation entailment, searches for contrary evidence, tests whether calculations reconcile and identifies assumptions that are doing material work. An integrator then prepares the memorandum with sections for observations, interpretation, scenarios, uncertainties and questions for the analyst. It cannot erase unresolved challenge items.

The analyst reviews the integrated memorandum. Approval records which claims were accepted, modified or rejected. Approved facts can be promoted to semantic memory with scope and expiry; tentative interpretations remain in the case. Corrections are classified by cause so that the institution knows whether to change a source rule, a skill, a calculation or a model instruction.

This organization has common dependencies. All roles rely on the issuer identity and observation date, so those fields are protected. Several agents use the document repository, creating a shared source dependency. The integrator depends on every specialist's completion status. The challenge role should not depend entirely on the integrator's summary because that would weaken independence. Mapping these dependencies reveals where validation, locking, versioning and timeouts are required.

The design can be tested role by role and end to end. Component tests ask whether each agent performs its contract. Interface tests introduce missing or malformed hand-offs. Scenario tests measure the complete memorandum on ordinary and exceptional cases. Economic tests compare useful completion with the current analyst process. The organization earns its complexity only if specialization or verification improves the result enough to justify coordination and review.

Generative-agent research demonstrates how models, memory and planning can produce persistent, interactive behavior [5]. The institutional lesson is not that a research office should simulate human personalities. It is that persistent agents require explicit memory structures, schedules and interaction rules. A financial firm adds a further requirement: the responsibility and authority of each role must remain attributable even when coordination is dynamic.

11 The responsibility map as a management artifact

MAIN IDEA

A responsibility map reveals who performs each task, what evidence and tools they use, what they may change and who owns the consequence.

The practical output of Part I is a responsibility map. Rows contain the significant responsibilities in the workflow. Columns identify purpose, input, capability, permitted tools, memory, output, acceptance test, escalation and owner. A dependency view then shows which responsibilities consume the outputs or state of others.

The map exposes omissions. If a role produces a conclusion but no role verifies source freshness, freshness has been assumed rather than assigned. If every agent may update memory, ownership of institutional state is ambiguous. If an exception has no named destination, the automated process will eventually invent an informal one. If the business owner cannot explain why two roles are separate, the architecture may contain performative complexity.

The artifact also improves dialogue between business and technology teams. Business leaders can challenge whether the responsibilities reflect the real process. Engineers can identify schemas, tools and failure handling. Risk and compliance can locate controls at specific handoffs. Finance can estimate cost by responsibility. Model-risk teams can distinguish a change to a model from a change to the organization in which the model operates.

Most importantly, the responsibility map prepares the transition to authority. It tells us who or what performs the work, but not yet what consequences each role may create without approval. That second question requires a delegation map. Keeping the maps separate prevents capability from quietly acquiring authority simply because it sits inside a working workflow.

12 Design studio: build the smallest useful agent organization

MAIN IDEA

The best agent organization is the smallest arrangement whose roles produce measurable information, control or throughput for the assigned task.

The design studio begins with a monthly portfolio-risk commentary. The current process requires position data, risk measures, market events, mandate limits and explanations from several desks. The temptation is to create a team of specialist agents immediately. The discipline is to begin with the smallest organization that can perform the responsibility credibly.

Write the completion condition first. The output must explain material changes in portfolio risk, reconcile to approved measures, cite relevant events, distinguish observation from interpretation and reach the risk committee by a specified time. It remains a draft until an authorized owner approves it.

Now identify responsibilities that require distinct capabilities. Controlled data services obtain positions and risk measures. A calculation service verifies changes and contributions. A generative agent may synthesize the material. Add a market-evidence specialist only if general retrieval repeatedly misses relevant events. Add a challenge role only if it possesses an independent method, such as reconciliation to source data or a mandate check.

For every proposed agent, write a contract containing inputs, outputs, tools, memory, constraints, completion and escalation. Draw the handoffs. Mark shared dependencies such as portfolio identity, observation time and currency. Decide which fields are protected from downstream modification. If two roles cannot be distinguished by responsibility or evidence, combine them.

Introduce three disruptions. First, a risk measure arrives late. Second, two desks provide incompatible explanations. Third, the market-evidence agent retrieves confidential information that the receiving role is not permitted to see. The organization should preserve uncertainty, enforce data boundaries and escalate rather than manufacture a seamless narrative.

Estimate coordination cost: model calls, tool calls, elapsed time, integration failures and human review. Compare it with a one-agent architecture and the current manual process. The chosen organization should earn every specialist through measurable information, control or throughput. The deliverable is a responsibility map and a short explanation of why the design is neither more centralized nor more distributed than the task requires.

Closing synthesis: organized intelligence and the question of authority

MAIN IDEA

Agentic AI creates a designed organization of intelligence; the next governance question is how much discretion that organization should receive.

Part I of Lecture 2 has taken us from a model to an organization. The model contributes flexible intelligence, but a working system surrounds it with controlled sources and operations. Tools extend what can be done. Skills give recurring work a procedural form. Memory allows qualified information to persist. Agents exercise bounded discretion over responsibilities that cannot be reduced conveniently to fixed rules. Multi-agent arrangements distribute those responsibilities and create new requirements for coordination, evidence and disagreement.

The result is an artificial organization in a precise sense: a designed division of intelligent work. Its quality cannot be judged by the sophistication of the model alone. We must examine the allocation of responsibility, the quality of the handoffs and the controls that preserve meaning as work moves through the system.

Yet one decisive question remains unanswered. A research agent may be capable of drafting an alert, but may it send the alert? It may be capable of changing a record, but may it overwrite the approved version? It may be capable of initiating a transaction, but who has authorized that action and under what conditions?

Capability and authority are different. Part II begins with that distinction. Once intelligence has an organization, the institution must decide how much discretion to delegate to it. The subject is no longer simply agentic architecture. It is autonomy, governance and the design of accountable delegation.

References

MAIN IDEA

These sources connect agentic systems with artificial intelligence, administrative behavior, coordination theory and multi-agent organization.

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